14 CONTROL SYSTEM

These cards are used in NEW and RESTART problems if a control system is desired. They are also used to define the generic control components employed with the self-initialization option. Input can also be used to compute additional quantities from the normally computed quantities. These additional quantities can then be output in major and minor edits and plots.

Two different card types are available for entering control system data, but only one type can be used in a problem. The digits CCC or CCCC form the control variable number (i.e., control component number). The card format 205CCCNN allows 999 control variables, where CCC ranges from 001-999. The card format 205CCCCN allows 9999 control variables, where CCCC ranges from 1-9999.

If the self-initialization option is selected, the data cards described in Section 14.3.19, Section 14.3.20, and Section 14.3.21 must be included. If loop flow control is to be included, the data cards described in Section 14.3.19 must also be included.

14.1 Control Variable Card Type, 20500000

If this card is omitted, card type 205CCCNN is used. If this card is entered, either card format can be selected. This card cannot be entered on RESTART problems if control components exist from the restart problem, in which case the card format from the restart problem must be used.

W1(I) Enter 999 to select the 205CCCNN format or 9999 (4095 also allowed) to select the 205CCCCN format.

14.2 Control Component Type, 205CCC00 or 205CCCC0

One card must be entered for each of the generic control components when using the self-initialization option.

W1(A) Alphanumeric name. Enter a name descriptive of the component. This name will appear in the printed output along with the component number. A limit of 10 characters is allowed for CDC 7600 computers, and a limit of 8 characters is allowed for most other computers.

W2(A) Control component type. Enter one of the component names, SUM, MULT, DIV, DIFFRENI, DIFFREND, INTEGRAL, FUNCTION, STDFNCTN, DELAY, TRIPUNIT, TRIPDLAY, POWERI, POWERR, POWERX, PROP-INT, LAG, LEAD-LAG, CONSTANT, SHAFT, PUMPCTL, STEAMCTL, FEEDCTL DIGITAL, or the command, DELETE. If DELETE is entered, enter any alphanumeric word in Word 1 and zeros in Words 3-8. No other control system cards are needed when DELETE is entered.

W3(R) Scaling factor. For a CONSTANT component, this quantity is the constant value. No additional words are entered on this card, and cards 205CCC01-CCC09 or

205CCCC1-CCCC9 are not entered. For the PUMPCTL, STEAMCTL, or FEEDCTL components, this is the gain multiplier (G) for the output signal.

W4(R) Initial value.

W5(I) Initial value flag. 0 means no initial condition calculation and W4 is used as the initial condition; 1 means compute initial condition.

W6(I) Limiter control. Enter 0, or omit this and the following words if no limits on the control variable are to be imposed. Enter 1 if only a minimum limit is to be imposed, 2 if only a maximum limit is to be imposed, and enter 3 if both minimum and maximum limits are to be imposed.

W7(R) Minimum or maximum value. This word is the minimum or maximum value if only one limit is to be imposed or is the minimum value if both limits are to be imposed.

W8(R) Maximum value. This word is used if both limits are to be imposed.

W9(A) '1D' or '3D'. If not entered, this value is assumed as '1D'.

14.3 Control Component Data, 205CCC01-CCC98 or 205CCCC1-CCCC8

The format of these cards depends on the control component type. An equation is used to describe the processing by each component. The symbol Y represents the control variable defined by the component. The symbols A_j, j = 1,2,...,J, represent constants defined by the control component input data. The variables V_j, j = 1,2,...,J, represent any of the variables listed in the minor edit input description. Besides hydrodynamic component data, heat structure data, reactor kinetic data, etc., any of the control variables including the variable being defined may be specified. The symbol S is the scale factor (or G, the gain multiplier, for self-initialization control components) on card 205CCC00 or 205CCCC0. The variables V_j use the code's internal units (SI). To use British units, the user must convert from SI to British using the scale factor S (or the gain multiplier G) and the constants A_j.

14.3.1 Sum-Difference Component

This component is indicated by SUM in Word 2 of card 205CCC00 or 205CCCC0. The sum-difference component is defined by

$$Y = S(A_0 + A_1V_1 + A_2V_2 + \ldots + A_jV_j) \ .$$

W1(R) Constant A_0.

W2(R) Constant A_1.

W3(A) Alphanumeric name of variable request code for V_1.

W4(I) Integer name of the variable request code for V_1. At least four words that define a constant and one product term must be entered. Additional sets of three words corresponding to Words 2-4 can be entered for additional product terms up to twenty product terms. One or more cards may be used as desired. Card numbers need not be strictly consecutive. The sign of A_j determines addition or subtraction of the product terms.

14.3.2 Multiplier Component

This component is indicated by MULT in Word 2 of card 205CCC00 or 205CCCC0. The multiplier component is defined by

$$Y = SV_1V_2...V_j \ .$$

W1(A) Alphanumeric name of the variable request code for V_1.

W2(I) Integer name of the variable request code for V_1. At least two words must be entered. Additional pairs of words can be entered on this or additional cards to define additional factors. Card numbers need not be strictly consecutive.

14.3.3 Divide Component

This component is indicated by DIV in Word 2 of card 205CCC00 or 205CCCC0. The divide component is defined by

$$Y = \frac{S}{V_1} ???\mathrm{or}??? Y = \frac{SV_2}{V_1} \ .$$

Specifying two words on the card indicates the first form, and specifying four words on the card indicates the second form. Execution will terminate if a divide by 0 is attempted.

W1(A) Alphanumeric name of the variable request code for V_1.

W2(I) Integer name of the variable request code for V_1.

W3(A) Alphanumeric name of the variable request code for V_2.

W4(I) Integer name of the variable request code for V_2.

14.3.4 Differentiating Components

These components are indicated by DIFFRENI or DIFFREND in Word 2 of card 205CCC00 or 205CCCC0. The differentiating component is defined by

$$Y = S\frac{dV_1}{dt} .$$

This is evaluated by

$$Y = \frac{2S}{\Delta t}(V_1 - V_{10}) - Y_o \qquad \text{(DIFFERENI)}$$

$$Y = S\frac{(V_1 - V_{10})}{\Delta t} \qquad \text{(DIFFREND)}$$

where Δt is the time step, and V_{10} and Y_0 are values at the beginning of the time step. The numerical approximations for the DIFFRENI and INTEGRAL components are exact inverses of each other. However, an exact initial value is required to use the DIFFRENI component, and erroneous results are obtained if an exact initial value is not furnished. The DIFFREND component uses a simple difference approximation that is less accurate and is not consistent with the integration approximation, but does not require an initial value. For these reasons, use of DIFFRENI is not recommended.

Since differentiation, especially numerical differentiation, can introduce noise into the calculation, it should be avoided if possible. When using control components to solve differential equations, the equations can be arranged such that INTEGRAL components can handle all indicated derivatives except possibly those involving noncontrol variables.

W1(A) Alphanumeric name of variable request code for V_1.

W2(I) Integer name of variable request code for V_1.

14.3.5 Integrating Component

This component is indicated by INTEGRAL in Word 2 of card 205CCC00 or 205CCCC0. The integrating component is defined by

$$Y = S\int_0^t V_1 dt$$

or, in Laplace notation,

$$Y(s) = \frac{SV_1(s)}{s} \quad .$$

This is evaluated by

$$Y = Y_o + S \bullet (V_1 + V_{10})\frac{\Delta t}{2}$$

where Δt is the time step and Y_o and V_{10} are values at the beginning of the time step.

W1(A) Alphanumeric name of the variable request code for V_1.

W2(I) Integer name of the variable request code for V_1.

14.3.6 Functional Component

This component is indicated by FUNCTION in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$Y = S[FUNCTION (V_1)]$

where FUNCTION is defined by a general table. This allows the use of any function that is conveniently defined by a table lookup and linear interpolation procedure. The function component can also be used to set limiting values.

W1(A) Alphanumeric name of the variable request code for V_1.

W2(I) Integer name of the variable request code for V_1.

W3(I) General table number of the function.

14.3.7 Standard Function Component

This component is indicated by STDFNCTN in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$Y = S[FNCTN (V_1, V_2,...)]$

where FNCTN is ABS (absolute value), SQRT (square root), EXP (e raised to power), LOG (natural logarithm), SIN (sine), COS (cosine), TAN (tangent), ATAN (arc tangent), MIN (minimum value), or MAX (maximum value). All function types except MIN and MAX must have only one argument; MIN and MAX function types must have at least two arguments and may have up to twenty arguments. If the control variable being defined also appears in the argument list of MIN or MAX, the old time value is used in the comparison.

W1(A) FNCTN.

W2(A) Alphanumeric name of the variable request code for V_1.

W3(I) Integer name of the variable request code for V_1.

14.3.8 Delay Component

This component is indicated by DELAY in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = SV_1 (t - t_d)$$

where t is time and t_d is the delay time.

W1(A) Alphanumeric name of the variable request code for V_1.

W2(I) Integer name of the variable request code for V_1.

W3(R) Delay time, t_d (s).

W4(I) Number of hold positions. This quantity, h, must be > 0 and ≤ 100. This quantity determines the length of the table used to store past values of the quantity V_1. The maximum number of time-function pairs that can be stored is h + 2. The delay table time increment, d_{TM}, is $d_{TM} = \frac{t_d}{h}$. The delayed function is obtained by linear interpolation for $V_1(t - t_d)$ using the stored past history. As the problem is advanced in time, new time values are added to the table. Once the table is filled, new values replace values that are older than the delay time. There are no restrictions on t_dT or d_{TM} relative to the time steps on cards 2NN. When a change in advancement time is made at a restart, the time values in this table are changed to have time values as if the problem in the restart had run to the new advancement time.

14.3.9 Unit Trip Component

This component is indicated by TRIPUNIT in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = S \bullet U(\pm T_1)$$

where U is 0.0 if the trip, T_1, is false and is 1.0 if the trip is true. If the complement of T_1 is specified, U is 1.0 if the trip is false and 0.0 if the trip is true.

W1(I) Trip number. A minus sign may prefix the trip number to indicate that the complement of the trip is to be used.

14.3.10 Trip Delay Component

This component is indicated by TRIPDLAY in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = ST_{rptim}(T_1)$$

where T_{rptim} is the time the trip last turned true. If the trip is false, the value is -1.0; if the trip is true, the value is 0 or a positive number.

W1(I) Trip number, T_1.

14.3.11 Integer Power Component

This component is indicated by POWERI in Word2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = SV_1^I \ .$$

W1(A) Alphanumeric name of the variable request code for V_1.

W2(I) Integer name of the variable request code for V_1.

W3(I) I.

14.3.12 Real Power Component

This component is indicated by POWERR in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = SV_1^R \ .$$

W1(A) Alphanumeric name of the variable request code for V_1.

W2(I) Integer name of the variable request code for V_1.

W3(R) R.

14.3.13 Variable Power Component

This component is indicated by POWERX in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = SV_1^{V_2} .$$

W1(A) Alphanumeric name of the variable request code for V_1.

W2(I) Integer name of the variable request code for V_1.

W3(A) Alphanumeric name of the variable request code for V_2.

W4(I) Integer name of the variable request code for V_2.

14.3.14 Proportional-Integral Component

This component is indicated by PROP-INT in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = S\left[A_1 V_1 + A_2 \int_0^t V_1 dt\right]$$

or in Laplace transform notation,

$$Y(s) = S\left[A_1 + \frac{A_2}{s}\right] V_1(s) .$$

If the control variable is initialized,

$$Y(t_0) = SA_1 V_1(t_0) .$$

If it is desired that the output quantity Y remain constant as long as the input quantity remains constant, V_1 must initially be 0 regardless of the initialization flag.

W1(R) A_1.

W2(R) A_2.

W3(A) Alphanumeric name of the variable request code for V_1.

W4(I) Integer name of the variable request code for V_1.

14.3.15 Lag Component

This component is indicated by LAG in Word 2 of card 205CCC00 or 205CCCC0. This component is defined by

$$Y = \int_0^t \left(\frac{SV_1 - Y}{A_1} \right) dt$$

or, in Laplace transform notation,

$$Y(s) = \frac{S}{1 + A_1 s} V_1(s) \quad .$$

If the control variable is initialized,

$$Y\,(T_0) = SV_1(t_0) \quad .$$

If the initialization flag is set on and if the initial values of Y and V_1 satisfy a specified relationship, Y remains constant as long as V_1 retains its initial value.

W1(R) Lag time, A_1 (s).

W2(A) Alphanumeric name of the variable request code for V_1.

W3(I) Integer name of the variable request code for V_1.

14.3.16 Lead-Lag Component

This component is indicated by LEAD-LAG in Word 2 of card 205CCC00 or 205CCCC0. The component is defined by

$$Y = \frac{A_1 SV_1}{A_2} + \int_0^t \left(\frac{SV_1 - Y}{A_2} \right) dt$$

or, in Laplace transform notation,

$$Y(s) = S \frac{1 + A_1 s}{1 + A_2 s} V_1(s) \quad .$$

If the control variable is initialized,

$Y(t_0) = SV_1(t_0)$.

If the initialization flag is set on and if the initial values of Y and V_1 satisfy a specified relationship, Y remains constant as long as V_1 retains its initial value.

W1(R) Lead time, A_1 (s).

W2(R) Lag time, A_1 (s).

W3(A) Alphanumeric name of the variable request code for V_1.

W4(I) Integer name of the variable request code for V_1.

14.3.17 Constant Component

Cards 205CCC01-CCC09 or 205CCCC1-CCCC9 are not entered. The quantity in Word 3 of card 205CCC00 or 205CCCC0 is the constant value used for this component.

14.3.18 Shaft Component

This component is indicated by SHAFT in Word 2 of card 205CCC00 or 205CCCC0. A GENERATR component may optionally be associated with a SHAFT component. The SHAFT component advances the rotational velocity equation

$$\sum_i I_i \frac{d\omega}{dt} = \sum_i \tau_i - \sum_i f_i \omega + \tau_c$$

where I_i is the moment of inertia of component i, ω is rotational velocity, τ_i is torque of component i, f_i is the friction factor of component i, and τ_c is an optional torque from a control component. The summations include the shaft as well as the pump, turbine, and generator components that are connected to the shaft.

The SHAFT control component differs somewhat from other control components. The scale factor on card 205CCC00 or 205CCCC0 must be 1.0. The initial value and optional minimum and maximum values have units (rad/s, rev/min), and British-SI units conversion are applied to these quantities. The output of the SHAFT in minor and major edits is in the requested units. Card number ranges are restricted so that both data to complete the SHAFT component description and optional data to describe a generator can be entered. Units conversion is applied to the following cards.

14.3.18.1 Shaft Description, 205CCC01-CCC05 or 205CCCC1-CCCC5.

W1(I) Torque control variable number. If 0, there is no contribution to torque from the control system. If nonzero, the control variable with this number is assumed to be a torque and is added to the torques from the other components attached to the shaft. The torque must be in SI units.

W2(R) Shaft moment of inertia, I_i (kg•m^2, lb•ft^2).

W3(R) Friction factor for the shaft, f_i (N•m•s, lb$_f$•ft•s).

W4(A) Type of attached component. Enter either TURBINE, PUMP, or GENERATR.

W5(I) Component number. This is the hydrodynamic component number for a TURBINE or PUMP, or the control variable number for this SHAFT component if GENERATR.

Additional two-word pairs may be entered to attach additional components to the shaft, up to a total of ten components. Only one generator, the one which is defined as part of this SHAFT component, may be attached.

14.3.18.2 Generator Description, 205CCC06 or 205CCCC6. Each SHAFT component may optionally define an associated GENERATR component.

W1(R) Initial rotational velocity (rad/s, rev/min).

W2(R) Synchronous rotational velocity (rad/s, rev/min).

W3(R) Moment of inertia, I_i (kg•m^2, lb•ft^2).

W4(R) Friction factor, f_i (N•m•s, lb$_f$•ft•s).

W5(I) Generator trip number. When the trip is false, the generator is connected to an electrical distribution system and rotational velocity is forced to the synchronous speed. When the trip is true, the generator is not connected to an electrical system and the generator and shaft rotational velocity is computed from the rotational velocity equation.

W6(I) Generator disconnect trip number. If 0, the generator is always connected to the shaft. If nonzero, the generator is connected to the shaft when the trip is false and disconnected when the trip is true.

14.3.19 PUMPCTL Component

This component is specified when using the self-initialization option and loop flow control is desired, but it is not limited to that use. For each PUMPCTL component enter:

W1(A) Alphanumeric name of setpoint variable.

W2(I) Parameter part of setpoint variable.

W3(A) Alphanumeric name of sensed variable.

W4(I) Parameter part of sensed variable.

W5(R) Scale factor(s) applied to sensed and setpoint values, S_i. Must be nonzero.

W6(R) Integral name time constant, T_2 (s).

W7(R) Proportional part time constant, T_1 (s).

Standard use of PUMPCTL controller require the following interpretation of the input data. W1 and W2 contain CNTRLVAR and CCC (or CCCC), respectively, where CCC (or CCCC) is a CONSTANT type control element containing the desired (setpoint) flow rate. W3 is MFLOWJ, and W4 is the junction number at which the flow is to be sensed and compared to the setpoint. W5 is the S_i value used to divide the difference between the desired (setpoint) and sensed flow rate to produce the error signal E_1. E_1 must be initially 0 if it is intended to have the controller output remain constant as long as the input quantities remain constant. W6 and W7 are the T_2 and T_1 values, respectively. All variables having units must be in SI units.

14.3.20 STEAMCTL Component

This component is specified when using the self-initialization option to control steam flow from one or more steam generators, but it is not limited to that use. For each STEAMCTL component enter:

W1(A) Alphanumeric name of setpoint variable.

W2(I) Parameter part of setpoint variable.

W3(A) Alphanumeric name of sensed variable.

W4(I) Parameter part of sensed variable.

W5(R) Scale factor(s) applied to sensed and setpoint values, S_j. Must be nonzero.

W6(R) Integral name time constant, T_4 (s).

W7(R) Proportional part time constant, T_3 (s).

Standard use of the STEAMCTL controller requires the following interpretation of the input data. W1 and W2 would contain CNTRLVAR and CCC (or CCCC), respectively, where CCC (or CCCC) is a

CONSTANT type control element. This constant would be the desired (setpoint) cold leg temperature (for suboptions A and B) or secondary pressure (suboptions C and D). W3 would be TEMPF (for suboptions A and B) or P (for suboptions C and D), and W4 would be the volume number where the temperature (suboptions A and B) or pressure (suboptions C and D) is sensed. W5 is the S_j value used to divide the difference between the desired (setpoint) and sensed temperature (suboptions A and B) or pressure (suboptions C and D) to produce the error signal E_2. E_2 must be initially 0 if it is intended to have the controller output remain constant as long as the input quantities remain constant. W6 and W7 are the T_4 and T_3 values respectively. All variables having units must be in SI units.

14.3.21 FEEDCTL Component

This component is specified when using the self-initialization option to control feedwater flow to a steam generator, but it is not limited to that use. For each FEEDCTL component enter:

W1(A) Alphanumeric name of first setpoint variable.

W2(I) Parameter part of first setpoint variable.

W3(A) Alphanumeric name of sensed variable to be compared with first setpoint.

W4(I) Parameter part of sensed variable to be compared with first setpoint.

W5(R) Scale factor applied to sensed and setpoint values (first setpoint), S_k. Must be nonzero.

W6(A) Alphanumeric name of second setpoint variable.

W7(I) Parameter part of second setpoint variable.

W8(A) Alphanumeric name of sensed variable to be compared with second setpoint.

W9(I) Parameter part of sensed variable to be compared with second setpoint.

W10(R) Scale factor applied to sensed and setpoint values (second setpoint), S_m. Must be nonzero.

W11(R) Integral name time constant, T_6 (s).

W12(R) Proportional part time constant, T_5 (s).

Standard use of the FEEDCTL controller requires the following interpretation of the input data. W1 and W2 contain CNTRLVAR and CCC (or CCCC), respectively, where CCC (or CCCC) is a CONSTANT type control element. This constant would be the desired (setpoint) steam generator secondary side water level. The latter may be expressed alternatively as a desired secondary coolant mass or as a differential pressure measured between two locations in the steam generator downcomer. W3 and W4 would contain CNTRLVAR and CCC (or CCCC), respectively, where CCC (or CCCC) is the number of the control component that describes the summing algorithm to compute the sensed variable (e.g., collapsed water level may be computed by summing the product of VOIDF and volume length over the control volumes in the riser section). W5 is the S_k value used to divide the difference between the desired (setpoint) and sensed water level to produce the first portion of the error signal E_3. W6 is MFLOWJ, and W7 is the junction number of the steam exit junction from the steam generator. W8 is MFLOWJ, and W9 is the junction number of the feedwater inlet junction. W10 is the S_m value used to divide the difference between the sensed steam flow and sensed feedwater flow to produce the second portion of the error signal E_3. E_3 must be initially 0 if it is intended to have the controller output remain constant as long as the input quantities remain constant. W11 and W12 are the T_6 and T_5 values, respectively. All variables having units must be in SI units.

14.3.22 DIGITAL Component

This component is indicated by DIGITAL in Word 2 of Card 205CCC00 or 205CCCC0. The component is defined by

$Y = SV_1 f\{t_s\}(t - t_d)$

where t is time and t_d is the delay time.

W1(A)	Alphanumeric name of the variable request code for V_1.
W2(I)	Integer name of the variable request code for V_1.
W3(R)	Sampling time, t_s (s).
W4(R)	Delay time, t_d (s).